World Financial Cycles and Global Trade

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The Question

- Emerging market spreads rise and fall together: the Global Financial Cycle
- Usual suspects are financial: US monetary policy, global risk appetite
- Miranda-Agrippino and Rey (2022): the second global factor tracks global trade
- **Can global trade shocks generate a world financial cycle?**

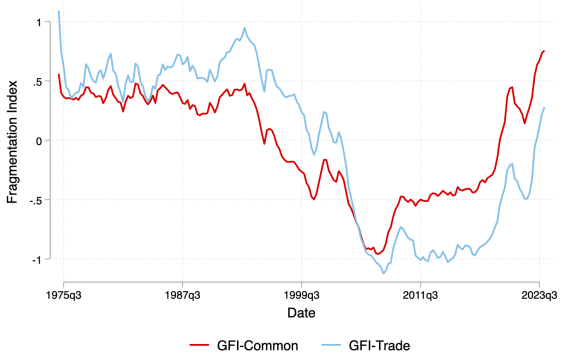
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- Two facts to explain: when global trade cost rises, spreads
  - *rise* on average and *become more dispersed* across countries

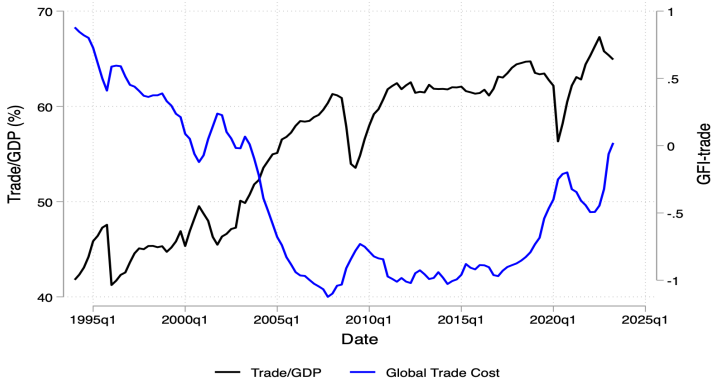
- **Common:** costlier imported inputs everywhere, spreads rise together
  - Under balanced trade, this is the whole story: equal losses, no change in dispersion
- **Differential:** *high debt countries save*, deeper recession, depreciate more
  - High debtors' spreads rise by more, dispersion widens

Geopolitical Fragmentation Index



- Trade cost proxied with quarterly *Geopolitical Fragmentation Trade Index* (GFI-trade)
  - Fernandez-Villaverde et al. (2024): dynamic hierarchical factor model
  - 88 countries data $\rightarrow$ 16 global variables, 4 categories

Global Trade and Trade Cost

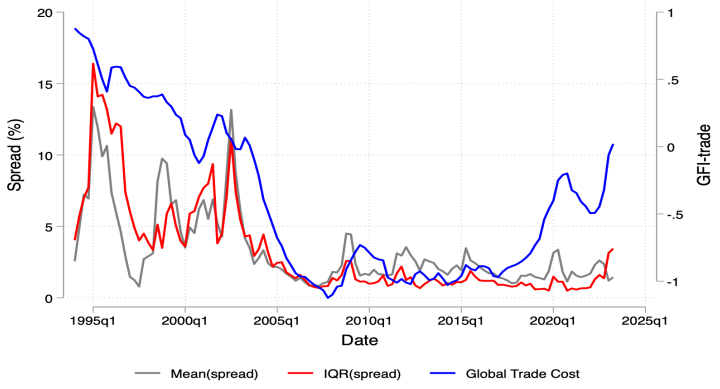


Falling global trade cost accompanied by larger trade flows

Global trade cost measured by GFI-Trade index from Fernández-Villaverde, Mineyama and Song (2025)

35 countries including small OECD + emerging countries

Global Trade Cost and Spread



Falling global trade cost accompanied by lower & less dispersed sovereign spreads

How do global trade shocks affect countries differentially?

This Paper

- Global trade barriers in emerging markets with unbalanced trade and default risk
- **Quantitatively:** *world GE model*, one advanced economy, many emerging markets
  - Gross trade and sovereign default with endogenous borrowing terms
- **Empirically:** *heterogeneous effect panel local projection*, 36 countries
  - Higher global trade cost: high debt countries lose more output, spreads rise more

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- **General equilibrium is central:** a global shock hits all emerging markets together
  - They cut imports and try to save at once: global demand and the world interest rate move
  - This loops back to the advanced economy, then to emerging markets via borrowing costs
  - *A world financial cycle driven by trade: spreads rise together and fan out*
- **Other experiments and counterfactuals:**
  - Advanced economies' tariffs, export subsidies, transitory vs permanent shocks, ...

A Model of Global Trade and Financial Cycle

World general equilibrium model: global trade + sovereign default

- Rep. advanced economy (AE) and a continuum of emerging markets (SOE)
- Gross trade flows: all import final goods from all, use as intermediates
- Sovereign default setting for SOEs: incomplete markets, default
- Global + idiosyncratic iceberg trade costs

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Mechanism: Following a *global trade cost shock* $\uparrow$

- Intermediate goods $\downarrow \Rightarrow$ output $\downarrow \Rightarrow$ sovereign default risk $\uparrow$ (**same across countries**)

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Mechanism: Following a *global trade cost shock* ↑

- Intermediate goods ↓ ⇒ output ↓ ⇒ sovereign default risk ↑ (**same across countries**)
- *High-debt countries* need to save: imports fall more than export
  ⇒ Greater output drop, greater depreciation
  ⇒ Disincentivize debt repayment
  ⇒ Sovereign default risk ↑ more (**heterogeneous across countries**)

Preview of Results

- Parameterize using cross-sectional/business cycle financial & trade moments
- Model results consistent with data:
  - Panel local projection with heterogeneous effects
  - Global trade costs $\uparrow \Rightarrow$ high-debt countries: greater output $\downarrow$, greater spread $\uparrow$
- Counterfactual: Stand-in advanced economy imposes tariffs on SOEs
  $\Rightarrow$ AE larger deficit, higher SOE spreads, larger global imbalances

- **Sovereign default.** Eaton and Gersovitz (1981); Aguiar and Gopinath (2006); Arellano (2008); Cuadra et al.(2010); Yue (2010); Hatchondo et al.(2016); Chatterjee and Eyigungor (2012, 2015); Na et al. (2018), Aguiar et al. (2016), Aguiar and Amador (2023), etc.
  Introduce trade frictions as a driver of dynamics in sovereign spreads

- **Global financial cycles:** Longstaff et al. (2011); Miranda-Agrippino and Rey (2020, 2022); Morelli et al. (2022); Gilchrist et al. (2022); Bai et al. (2024), etc.
  Aggregate trade shocks impact trend and time-varying dispersion of spreads

- **World G.E. models with int'l financial markets**: Bai and Zhang (2012); Morelli et al. (2022); Alessandria et al. (2024), etc.
  World GE model of interactions between gross trade and limited enforcement

- **Trade costs consequences of sov defaults:** Rose (2005); Borensztein and Panizza (2010); Martinez and Sandleris (2011); Mendoza and Yue (2012); Zymek (2012); Serfaty (2021); Andreasen et al. (2024); Alessandria et al. (2024), etc.
  Impact of sovereign defaults on trade, through global equilibrium effects

A Model of Gross Trade Flows and Default

Model

- World general equilibrium model:
  - Measure ξ continuum of small open economies (SOE), discount factor β
  - One stand-in advanced economy (AE), discount factor $\beta_{AE} > \beta$
- Countries
  - Produce using labor L, domestic H and imported intermediate good M
  - Iceberg trade costs $\tau_t \epsilon_{it} p_t^M M_{it}$
- Imperfect international financial markets
  - Centralized borrowing and default decision
  - Sovereign default $\Rightarrow$ haircut + productivity loss (no market exclusion)
- Idiosyncratic productivity shock z_{it}, trade cost shock ϵ_{it} + global trade cost shock τ_t

Individual Country: Production

- Each country i produces a tradable differentiated good with labor, domestic intermediate H, and imported intermediate M

$$Q_{it} = z_{it} L_{it}^{\alpha} G_{it}^{\theta},$$

where

$$G_{it} = \left(\omega H_{it}^{\frac{\gamma-1}{\gamma}} + (1-\omega) M_{it}^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}$$

- Firms take prices as given and solve

$$\max_{L_{it}, H_{it}, M_{it}} \left\{ p_{it} Q_{it} - w_{it} L_{it} - p_{it} H_{it} - p_t^M \tau_t M_{it} \right\},$$

Individual Country: Households

- A representative household's optimization problem

$$\max_{\{C_{it},L_{it}\}} \mathbb{E}\sum_{t=0}^{\infty}\beta^t \frac{\left(C_{it}-\chi\frac{L_{it}^{1+1/\nu}}{1+1/\nu}\right)^{1-\sigma}}{1-\sigma}$$

$$\text{s.t.} \quad p_{it}C_{it} = w_{it}L_{it} + T_{it} + \pi_{it},$$

  - T_{it}: lump sum tax or transfer from government
  - π_{it}: profits of the representative firm

- Optimal GHH labor supply: $\chi L_{it}^{1/\nu} = \frac{w_{it}}{p_{it}}$

Individual Country (SOE): Sovereign

- Country i issues claims to b_{it+1} units of long-term defaultable bonds at t
  - Sequence of payments starting from $t+1$: κ, $\kappa(1-\delta)$, $\kappa(1-\delta)^2$,...
  - δ: controls bond duration

- The sovereign decides whether to default and future bond position b_{it+1}
  - Default ($d_{it}=1$) $\Rightarrow$ *immediate* debt reduction, $b_{it} \to \phi b_{it}$, with $\phi < 1$
  - Cost: utility cost (ψ) + productivity loss ($\tilde{z}_{it} = h(z_{it}, d_{it}) \leq z_{it}$)

- Budget constraint

$$T_{it} = -\kappa \phi^{d_{it}} b_{it} + q_t(z_{it}, b_{it+1}) \left[b_{it+1} - (1-\delta)\phi^{d_{it}} b_{it} \right]$$

Advanced Economy

- Identical production structure as SOEs
- No idiosyncratic shocks
- No default, bond economy
- Later, AE import tariffs (+ export tax or subsidy in paper, other experiments)

Global Intermediate Producer

- Competitive global intermediaries assemble goods from all countries:

$$Y_t = \left(X_{\text{AE},t}^{\frac{\eta-1}{\eta}} + \xi \int_0^1 X_{it}^{\frac{\eta-1}{\eta}} di \right)^{\frac{\eta}{\eta-1}}$$

- Optimization

$$\max_{X_{\text{AE},t},\{X_{it}\}} p_t^M Y_t - \xi \tau_t \int_0^1 p_{it} \epsilon_{it} X_{it} di - \underbrace{p_{\text{AE},t}}_{=1} \tau_t X_{\text{AE},t}$$

$\Rightarrow$ demand function and price index for global intermediate

$$X_{AE,t} = \left(\frac{\tau_t}{p_t^M} \right)^{-\frac{1}{\eta}} Y_t, \qquad X_{it} = \left(\frac{\tau_t \epsilon_{it} p_{it}}{p_t^M} \right)^{-\frac{1}{\eta}} Y_t, \qquad p_t^M = \tau_t \left(1 + \xi \int_0^1 (\epsilon_{it} p_{it})^{1-\eta} di \right)^{\frac{1}{1-\eta}}$$

Static Private Trade Equilibrium

Given the trade cost τ_t and distribution $\Omega_t(\tilde{z}_{it}, \epsilon_{it}, T_{it})$, the static-private equilibrium allocations $\{C_{it}, L_{it}, Q_{it}, X_{it}, M_{it}\}$, prices $\{w_{it}, p_{it}, p_t^M\}$ satisfy the following

- Firms' optimization, for each i and AE
- Households optimizes, for each i
- Global intermediate producers optimize
- Goods market clearing and balance of payment, for each i

$$Q_{it} = C_{it} + H_{it} + \tau_t \epsilon_{it} X_{it}, \qquad T_{it} + \underbrace{p_{it}\tau_t \epsilon_{it} X_{it}}_{\text{Exports}} = \underbrace{p_t^M \tau_t \epsilon_{it} M_{it}}_{\text{Imports}}$$

- Global intermediate goods clearing, $Y_t = \tau_t \left(M_{\text{AE},t} + \xi \int_0^1 \epsilon_{it} M_{it} di \right)$

Dynamic World Equilibrium: AE

- AE: trades assets internationally, full commitment
- Euler equation

$$u_{\mathrm{AE},t} q_t^{\mathrm{rf}} = \beta_{\mathrm{AE}} u_{\mathrm{AE},t+1} \left(\kappa + (1-\delta) q_{t+1}^{\mathrm{rf}} \right)$$

- Define short-term risk-free rate

$$R_t = \frac{1}{\beta_{\mathrm{AE}}} \frac{u_{\mathrm{AE},t}}{u_{\mathrm{AE},t+1}}$$

Dynamic World Equilibrium: SOE

Given aggregate state variable $S_t = (\tau_t, \Omega_t(z_{it}, \epsilon_{it}, b_{it}))$, government i solves

- Default decision, at the start of the period,

$$V_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t) = \max_{d_{it} \in \{0,1\}} \{d_{it} [W_{it}(\tilde{z}_{it}, \epsilon_{it}, \phi b_{it}, S_t) - \psi] + (1 - d_{it}) W_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t)\}$$

- Then, new asset position

$$\begin{aligned} W_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t) = \max_{b_{it+1}} & \{u(C_{it}, L_{it}) + \beta \mathbb{E}_t V(z_{it+1}, \epsilon_{it+1}, b_{it+1}, S_{t+1})\} \\ \text{s.t. } & T_{it} = -\kappa b_{it} + q_t(z_{it}, b_{it+1}) [b_{it+1} - (1 - \delta) b_{it}], \\ & C_{it} = \mathcal{C}_{it}(z_{it}, \epsilon_{it}, T_{it}, \Omega_t, \tau_t), \quad L_{it} = \mathcal{L}_{it}(z_{it}, \epsilon_{it}, T_{it}, \Omega_t, \tau_t), \\ & \Omega_{t+1} = H_\Omega(\Omega_t, \tau_t) \end{aligned}$$

Bond Pricing

- Intl' lenders: risk neutral and competitive, fully committed
  - $b_{it+1} \leq 0$: Country i holds long-term bonds issued by international lenders
  - $b_{it+1} > 0$: Lenders hold long-term risky bonds issued by country i
- Introduce short-term, gross risk-free rate R_t
- Equilibrium bond prices:

$$q_{it} = \begin{cases} q_t^{rf}, & \text{if } b_{it+1} \leq 0 \\ \frac{1}{R_t} \mathbb{E}_t \left[\phi^{d_{it+1}} \left(\kappa + (1-\delta) q_{it+1} \right) \right], & \text{otherwise} \end{cases}$$

with $q_t^{rf} = \frac{1}{R_t} \left(\kappa + (1-\delta) q_{t+1}^{rf} \right)$

Market Clearing

- Goods market clearing and balance of payment, for each i

$$Q_{it} = C_{it} + H_{it} + \tau_t \epsilon_{it} X_{it},$$

$$\underbrace{p_{it}\tau_t\epsilon_{it}X_{it}}_{\text{Exports}} - \underbrace{p_t^M\tau_t\epsilon_{it}M_{it}}_{\text{Imports}} = \underbrace{\kappa\phi^{d_{it}}b_{it} - q_t(z_{it}, b_{it+1})\left[b_{it+1} - (1-\delta)\phi^{d_{it}}b_{it}\right]}_{\text{govt transfer } (-T_{it})}$$

- Global intermediate goods clearing,

$$Y_t = \tau_t \left(M_{\text{AE},t} + \xi \int_0^1 \epsilon_{it} M_{it} di \right)$$

- Bonds market clearing:

$$q_t^{\text{rf}} b_{\text{AE},t} + \xi \int q_t(z_{it}, \epsilon_{it}, b_{it+1}) b_{it+1} d\Omega_t = 0.$$

Differential Effects of Global Trade Shocks

- Intuition from static trade equilibrium, for fixed trade balances (T_{it})
- The Keynes–Ohlin "transfer problem"
- $T_{it} < 0$: countries who need to save (usually high-debt countries)

Static Private Equilibrium

Given $\{\tau, Y, p^M\}$ and (z, T) for each country, each SOE's $\{C, L, H, M, X, Q, p\}$ solve

$$X = \left(\frac{\tau p}{p^M}\right)^{-\eta} Y \qquad \text{(export demand)}$$

$$\alpha Q/L = \chi L^{1/\nu} \qquad \text{(labor market)}$$

$$\tau p^M/p = \theta(1-\omega) Q G^{\frac{1}{\gamma}-1} M^{-\frac{1}{\gamma}} \qquad (M \text{ demand})$$

$$1 = \theta\omega Q G^{\frac{1}{\gamma}-1} H^{-\frac{1}{\gamma}} \qquad (H \text{ demand})$$

$$Q = z L^{\alpha} G(H, M)^{\theta} \qquad \text{(gross output)}$$

$$C = Q - H - \tau X \qquad \text{(domestic resources)}$$

$$T + \tau p X = \tau p^M M \qquad \text{(BoP)}$$

Similarly for AE conditions, but $p_{AE} = 1$ and drop BoP
Global intermediate goods clearing $Y = \tau \left(M_{\text{AE}} + \xi \int M(z, T) d\Omega(z, T)\right)$

Mechanism: Effects of Trade Cost Shocks

Following a trade cost shock $\tau \uparrow$,

- Output Q, exports, imports $\downarrow$ for all countries (z, T)

Mechanism: Effects of Trade Cost Shocks

Following a trade cost shock $\tau \uparrow$,

- Output Q, exports, imports $\downarrow$ for all countries (z, T)
- Under balanced trade, $T = 0$, and all countries (SOEs & AE) have same $\%\triangle Q$
  - Independent of country size

Mechanism: Effects of Trade Cost Shocks

Following a trade cost shock $\tau \uparrow$,

- Output Q, exports, imports $\downarrow$ for all countries (z, T)
- Under differential trade balances,
  Greater saving incentives ($T < 0$) $\Rightarrow$ $Q \downarrow$ more, $p \downarrow$ (depreciate) more

$$\underbrace{-T/p}_{\text{saver}>0} = \tau X - \tau p^M M \approx \tau X - (1-\theta)Q$$

  - Imports (and thus Q) $\downarrow\downarrow$, exports $X = \left(\tau p / p^M\right)^{-\eta} Y$ fall by less (thus $p \downarrow\downarrow$)
  - Fall in p further increases real trade surplus $\Rightarrow$ *worse recession* Q

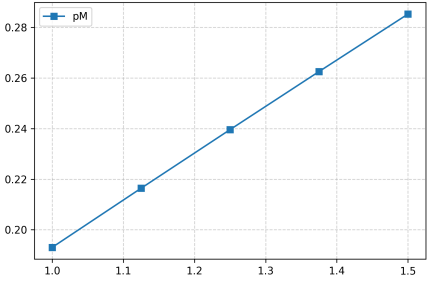
  - **Disincentivize gov't from repaying debt**

Simple Numerical Illustration

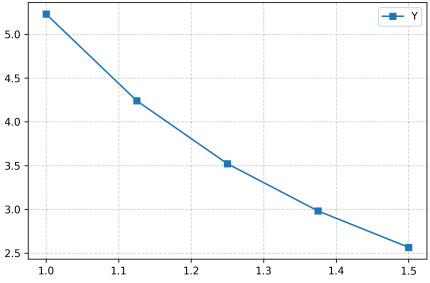
- Fix standard parameter values:
  $\alpha = 0.4, \theta = 0.43, \chi = 1.0, \nu = 0.72, \eta = 3, \gamma = 1, \omega = 0.6$

- Equilibria with $\tau \in [1, 1.5]$
  - Total transfer equals zero, $T_{AE} + \zeta \sum T_i = 0$
  - $T_i < 0$: net exporter (*larger saving incentives*)

Global Trade Shock: Average Effect



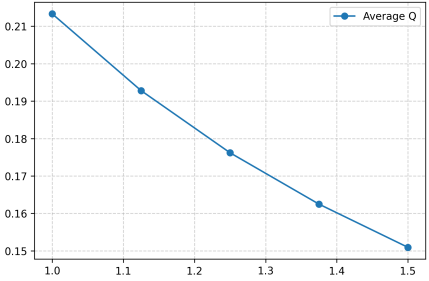
$\tau \Rightarrow p^M$



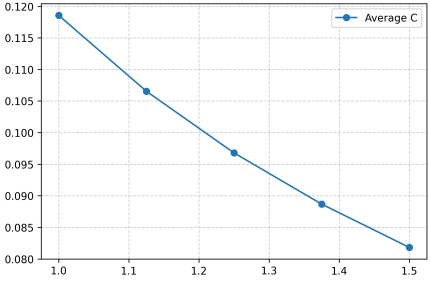
$\tau \Rightarrow Y$

- Higher trade cost $\Rightarrow$ increase in global intermediate price p^M, fall in global demand Y

Global Trade Shock: Average Effect



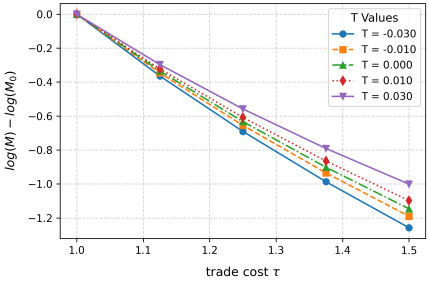
$\tau \Rightarrow Q$



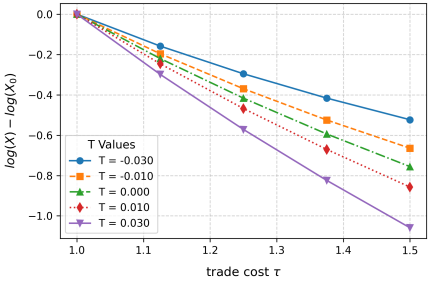
$\tau \Rightarrow C$

- Higher trade cost $\Rightarrow$ fall in output & consumption

Global Trade Shock: Differential Effect



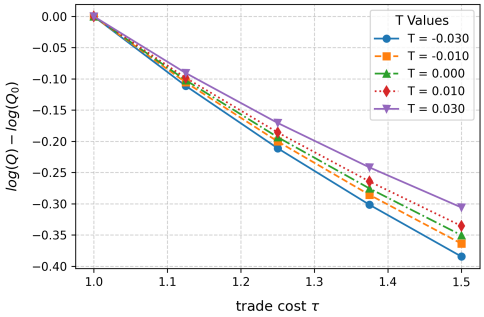
$\tau \Rightarrow M$



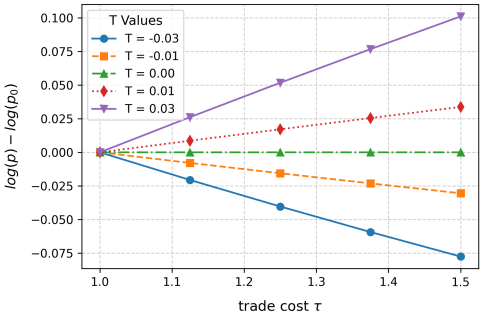
$\tau \Rightarrow X$

After $\tau \uparrow$, with grater savings $(T < 0) \Rightarrow$ *Imports falls by more, exports falls by less*

Global Trade Shock: Differential Effect



$\tau \Rightarrow Q$

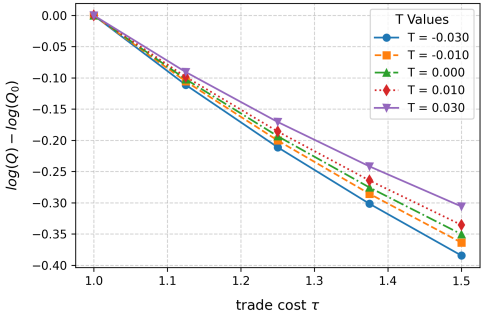


$\tau \Rightarrow p$

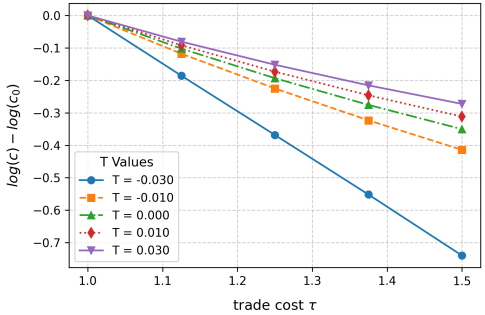
After $\tau \uparrow$, savings more ($T < 0$) $\Rightarrow$

- Q falls by more, p falls by more (greater depreciation)

Global Trade Shock: Differential Effect



$\tau \Rightarrow Q$



$\tau \Rightarrow C$

After $\tau \uparrow$, savings more ($T < 0$) $\Rightarrow$

- C falls by even more: $C = \theta Q + T/p$

Full Model, Dynamic General Equilibrium

- Gov't internalizes the impact of τ
- Endogenous borrowing and default decision
- Global bonds markets clear

Quantitative Analysis

- Parameterize using trade and sovereign spreads moments
  - At the invariant distribution, $R = 1/\beta_{AE}$
  - Data for AE: consolidate USA + Germany + Japan + China
  - Data SOE: 36 countries, OECD small + emerging markets

- Differential effect after global trade cost shock
  - Which country exhibits greater spreads increases?

- One-time "MIT" transitory shock experiments
  - Glocal trade cost τ, AE tariff, global v idiosyncratic trade costs, default v bond SOEs
  - More in the paper: transitory v permanent shocks, AE export tax or subsidy, ...

Parameterization

Assigned Parameters	
Risk aversion	$\sigma = 2$
Elas. of substitution:	$\gamma = 3$
Trade elasticity	$\eta = 3$
Labor share	$\alpha = 0.40$
Intermediate share	$\theta = 0.43$
Frisch elasticity	$\nu = 0.72$
Discount factor AE	$\beta = 0.995$
Debt duration	$\delta = 1/16$
Debt recovery rate	$\phi = 0.5$
Productivity persistence	$\rho_z = 0.9$
SOE trade cost persistence	$\rho_\epsilon = 0.9$

Parameters from Moment Matching	
Using moments from stationary dist.	
Disutility of working	$\chi = 441.5$
Import share of AE	$\omega_{AE} = 0.31$
Import share of SOE	$\omega = 0.16$
Mass of SOE	$\xi = 56.7$
AE mean productivity	$\bar{z}_{AE} = 32.8$
SOE mean productivity	$\bar{z}_{SOE} = 2.73$
Using moments in simulation	
Discount factor SOE	$\beta = 0.959$
Productivity volatility	$\sigma_z = 0.87\%$
SOE trade cost volatlity	$\sigma_\epsilon = 2.2\%$
Default costs	$\lambda_0 = -0.9$
	$\lambda_1 = 0.945$

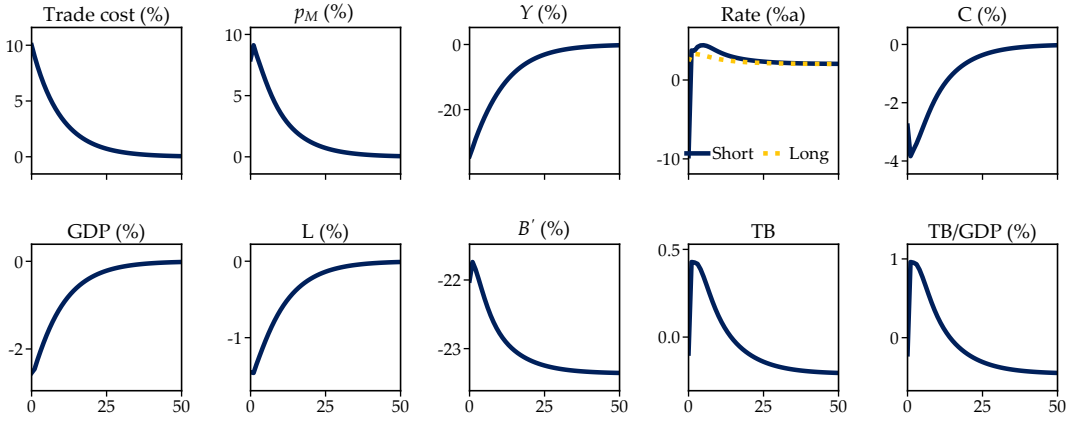
Targeted Moments

	Data	Model
Moments from Stationary Dist. (%)		
AE share of world GDP	49.5	49.5
Average SOE GDP over GDP_{AE}	1.8	1.8
Import share of GDP, AE	16.2	16.2
Import share of GDP, SOE	29.4	29.4
Business cycle Moments, SOE (%)		
Std. real GDP	5.2	6.0
Mean spread	1.0	2.1
Mean Debt to output	25.8	24.7
Std. spread	1.0	0.8
Std consumption/std GDP	1.0	1.0
Corr(spread, import/GDP)	-35	-38
Corr(export/gdp, import/GDP)	83	64

	Value	Internal	Source or Target
Production and Trade			
ζ	56.7	×	AE share of World GDP
z_{AE}	32.8		Normalization of Q_{AE}
z_{SOE}	2.73	×	Relative GDP size of average SOE to AE
ω_{AE}	0.31	×	Import share AE
ω	0.16	×	Import share SOE
α	0.4		Returns to scale, Labor
θ	0.43		Return to scale, Intermediates
η	3.0		Imported intermediates substitution
γ	3.0		Domestic-imported intermediates substitution
ρ_z	0.9		Arellano et al. 2025
ρ_ϵ	0.9	×	Corr spread, imports-to-GDP
σ_z	$8.7e^{-3}$	×	SOE GDP volatility
σ_ϵ	$2.2e^{-2}$	×	Corr imports, exports
Preferences			
σ	2.0		Reference value
β_{AE}	0.995		Short-term risk-free rate
β	0.959	×	Std log consumption to std log GDP ratio
χ	441.5		Normalization of AE labor supply
ν	0.72		Frisch elasticity, Healthcote et al. 2010
Debt and Default			
δ	16^{-1}		Macaulay duration
λ_0	−0.9	×	Spread
λ_1	0.945	×	Spread volatility
ϕ	0.5		Haircut, excl. preemptive, Cruces and Trebesch 2013
ψ	0.31	×	SOE debt-to-GDP
$\bar{q}$	0.6		Underwriting stds., Chatterjee and Eiygongur 2012

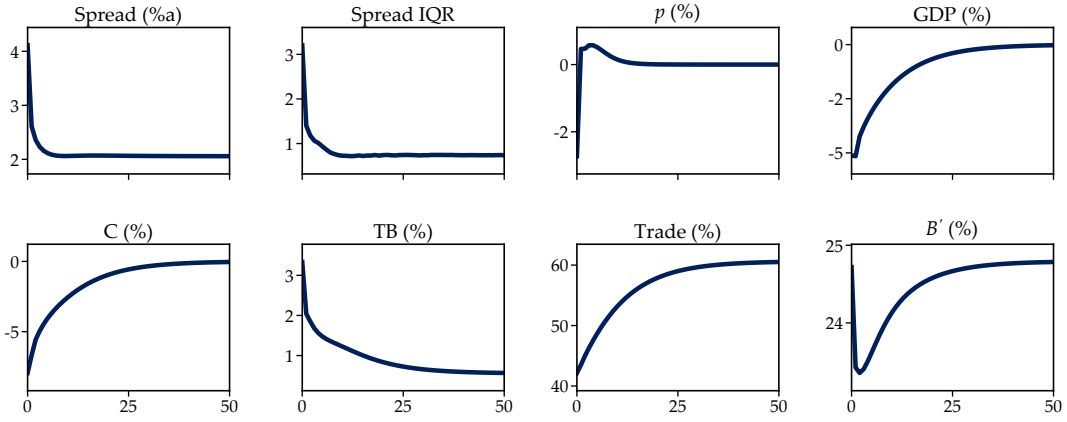
	Data	Model	Targeted
Means			
AE Share of World GDP	49.5	49.5	×
Average SOE to AE GDP	1.8	1.8	×
Spread	1.0	2.1	×
Debt-to-GDP	25.8	24.7	×
Imports-to-GDP AE	16.2	16.2	×
Imports-to-GDP SOE	29.4	30.0	×
Trade Balance-to-GDP	1.0	0.5	
Standard Deviations			
Log GDP	5.2	6.0	×
Spread	1.0	0.8	×
Std Log C / Std Log GDP	1.0	1.0	×
Trade Balance-to-GDP	3.5	2.3	
Exports-to-GDP	6.1	2.7	
Imports-to-GDP	6.0	2.8	
Correlations			
Spread, Log GDP	−11	−45	
Spread, Trade Balance-to-GDP	19	59	
Spread, Exports-to-GDP	−12	11	
Spread, Imports-to-GDP	−35	−38	×
Trade Balance-to-GDP, Log GDP	−18	24	
Exports-to-GDP, Log GDP	7	24	
Imports-to-GDP, Log GDP	19	3	
Exports, Imports	83	64	×

Global Trade Cost Shock: AE



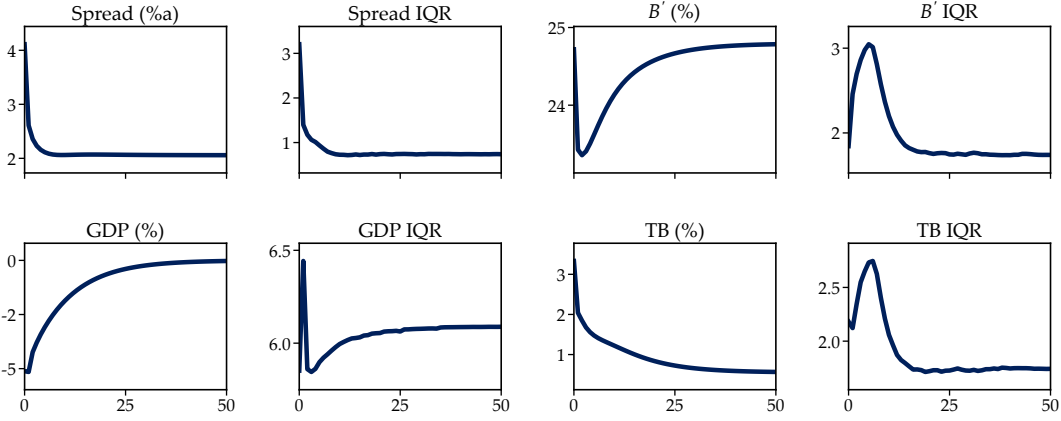
- Higher global trade shock $\tau \Rightarrow$ lower global intermediate production Y, raises price p^M
  lower AE GDP and consumption, higher interest rate

Global Trade Cost Shock: SOE



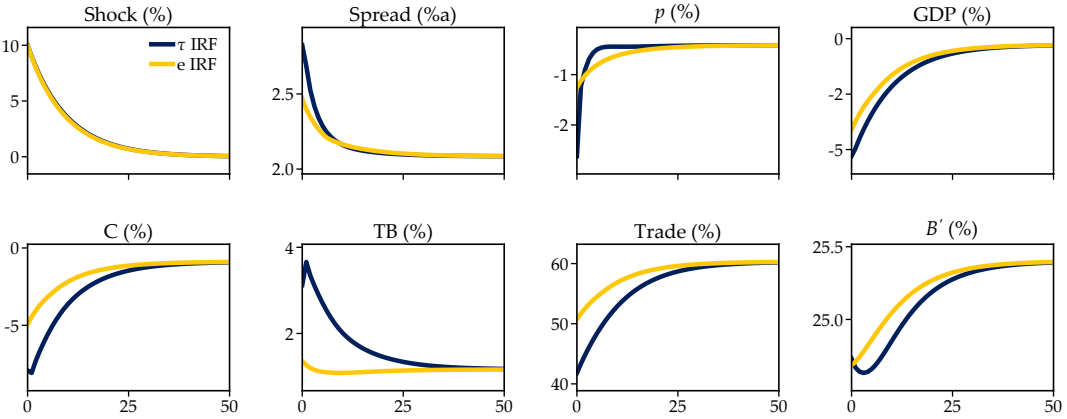
- Higher global trade shock $\tau \Rightarrow$ higher sovereign spreads, depreciation, lower GDP, trade surplus (TB$>$ 0) for SOE

Global Trade Cost Shock: SOE Dispersions



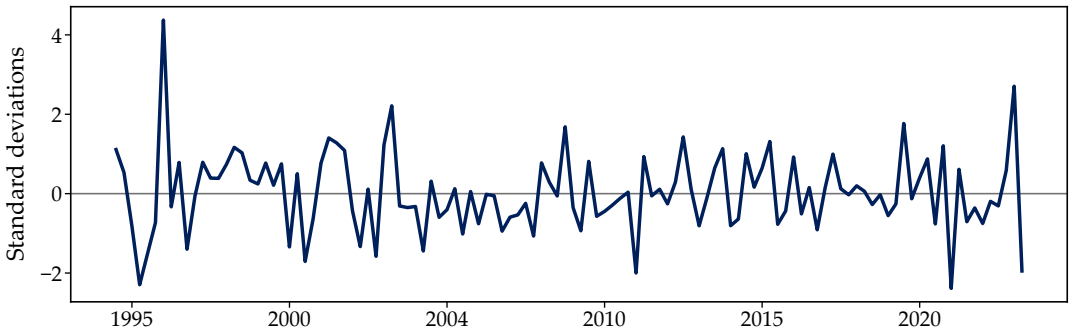
- Higher global trade shock $\tau \Rightarrow$ larger dispersion in spreads and trade balances among SOEs

Global τ versus Idiosyncratic e Trade Costs



- Same shock magnitude and persistence, both enter symmetrically in all equations.
  Difference: τ features *G.E. effects* via p^M, Y, and R

Trade-Fragmentation Shock



The regressor s_t is this SVAR innovation to GFI-trade, standardized to one standard deviation.

Testing the Differential Effect of τ

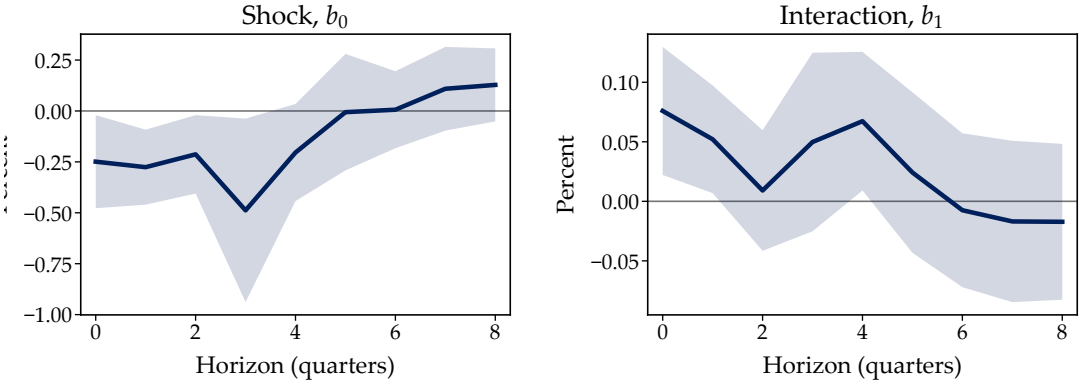
Unbalanced quarterly panel, 36 countries. At horizon $h = 0, \ldots, H$,

$$y_{i,t+h} = \alpha_i^h + b_0^h s_t + b_1^h (s_t \times z_{i,t-1}) + g^h z_{i,t-1} + \sum_{k=1}^{3} \phi_k^h \tilde{y}_{i,t-k} + u_{i,t+h}$$

Where

- s_t is the GFI-Trade SVAR innovation. A positive value is greater trade fragmentation.
- $z_{i,t-1}$ is NIIP/GDP, demeaned within quarter & normalized by its pooled s.d.
- b_0^h is the response at the mean position.
  b_1^h is the additional effect, per one-s.d. higher NIIP/GDP.
- y is HP-filtered log GDP or the log sovereign spread. $\tilde{y}$ is HP-filtered log GDP.

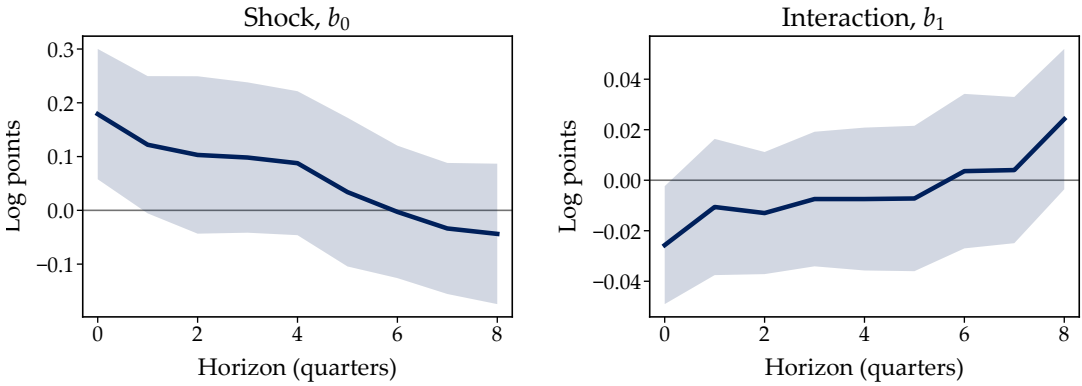
Response of Real GDP



Percent deviation from an HP trend. Bands are 90% confidence intervals.

- Lower NIIP (more indebted) $\Rightarrow$ deeper recession.

Response of Sovereign Spread



Log points. Bands are 90% confidence intervals.

- Lower NIIP (more indebted) ⇒ greater spread increase.

Towards an Analysis of AE Tariffs

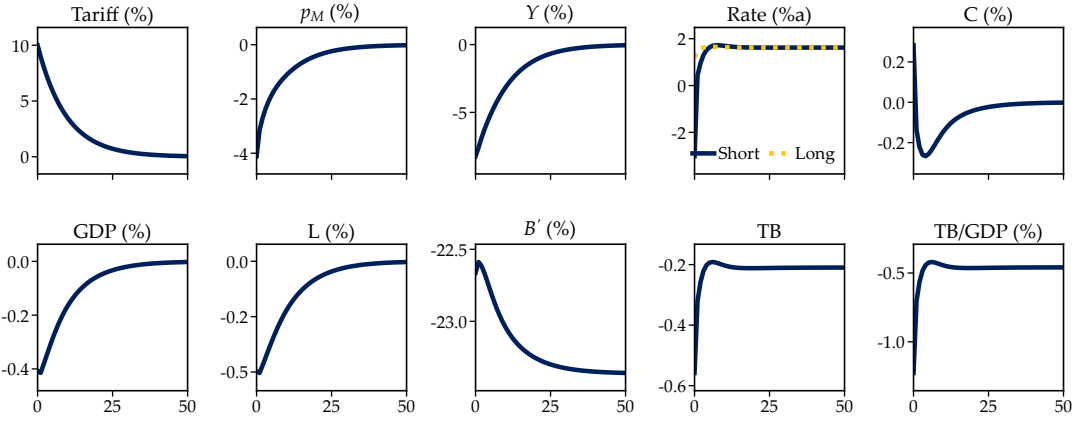
- Advanced economy imposes tariffs ε^m_{AE}
- Equilibrium conditions

$$\tau p^M (1 + \varepsilon^m_{AE}) = \theta(1-\omega) Q G^{\frac{1}{\gamma}-1} M^{-\frac{1}{\gamma}} \qquad (M \text{ demand})$$

Other conditions same as before, including

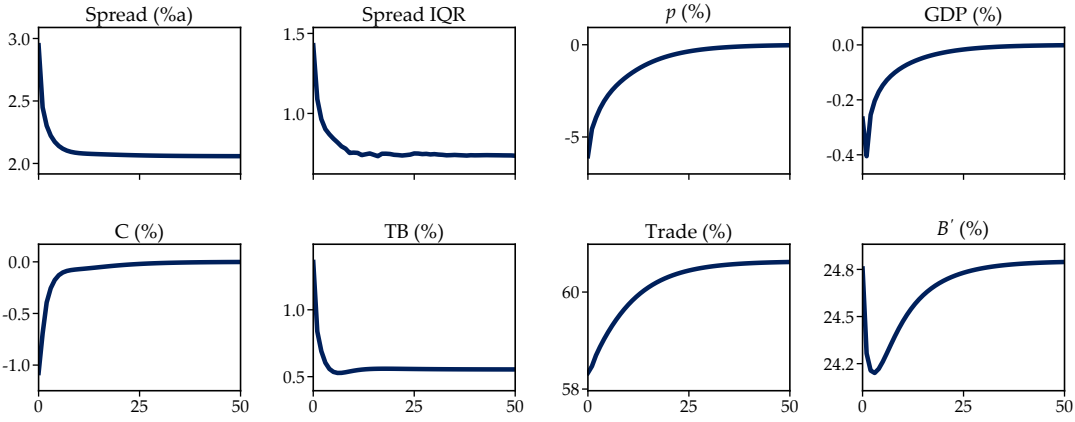
$$C_{AE} = Q_{AE} - H_{AE} - \tau X_{AE} \qquad \text{(domestic resources)}$$

AE Tariff: AE



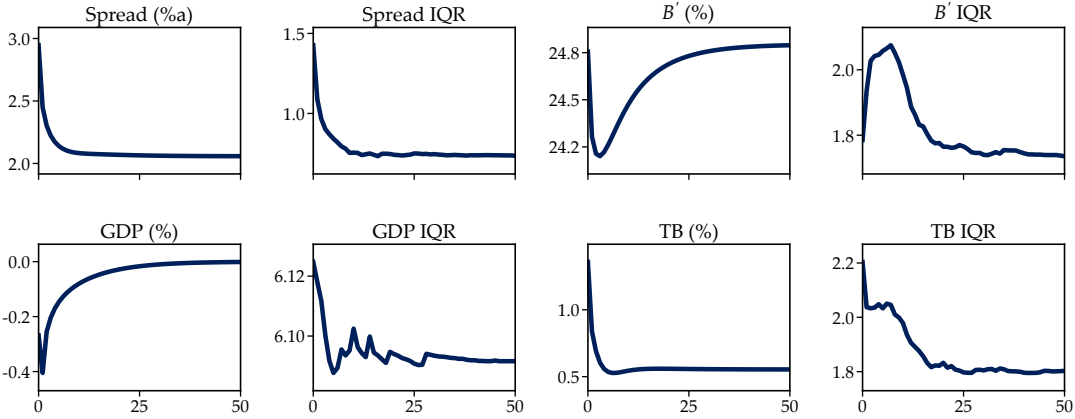
- Higher AE tariff $\Rightarrow$ lower global intermediate production Y, *lowers price* p^M
  raise AE consumption, *lower* interest rate

AE Tariff: SOE



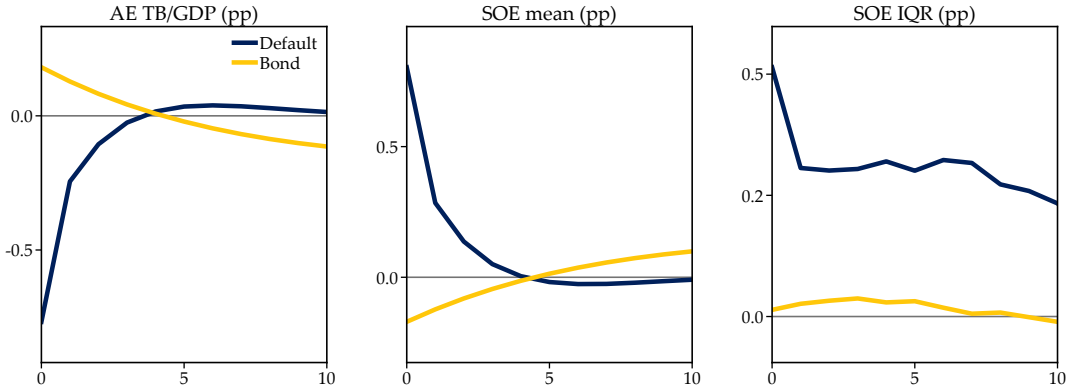
- Higher AE tariff $\Rightarrow$ higher sovereign spreads, depreciation,
  lower GDP, trade surplus (TB>0) for SOE
  Trade deficit for AE

AE Tariff: SOE Dispersions



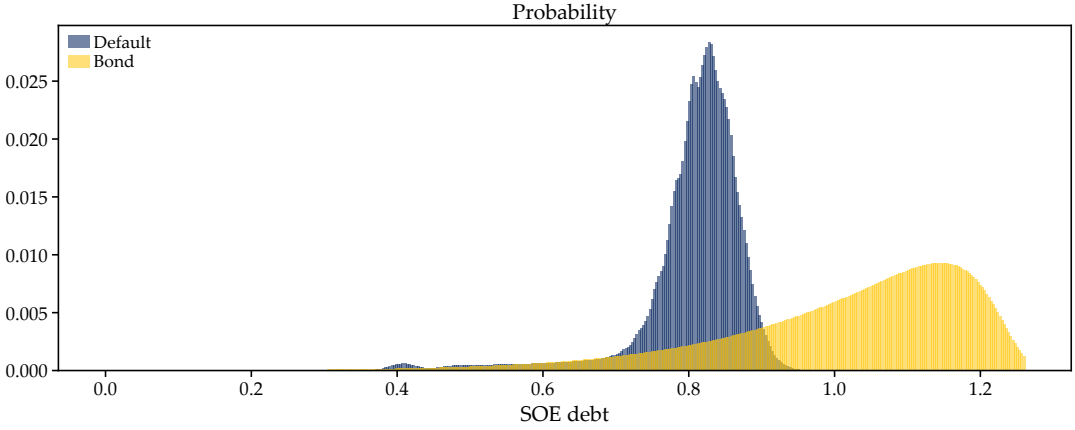
- Higher AE tariff $\Rightarrow$ larger dispersion in spreads, GDP, trade balances among SOEs

AE Tariff: Default versus Risk-free Bond Economies



- **Can tariffs increase the trade balance of the AE?** SOEs' *financial friction severity* is key. Tight borrowing schedule from default risk? Then no.

Default versus Risk-free Bond Economies: Debt Positions



- Risk-free bond economy with loose borrowing limit, plus patient SOEs with mass away from limit. Tariff hits, SOEs borrow and AE willing to save. With default, SOEs must export.

Conclusion

- We build a world GE model with gross trade and sovereign default
- We find differential effects of global trade cost shocks
  - High-debt countries (savers): greater fall in output & larger increase in spreads
  - Explains movements in average and cross-country dispersion of spreads
- A framework to study impact of tariffs from advanced economies under imperfect international financial markets

Appendix

Default and Bond Economy: Trade Cost

